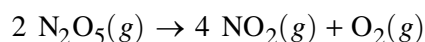


Unit 5: AP Free Response Practice #1 [8 points]
2022 FRQ #5

5. The following equation represents the decomposition of N_2O_5 , for which the rate law is $\text{rate} = k[\text{N}_2\text{O}_5]$.

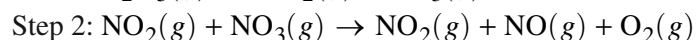
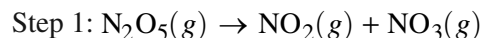


A sample of pure $\text{N}_2\text{O}_5(g)$ is placed in an evacuated container and allowed to decompose at a constant temperature of 300 K. The concentration of $\text{N}_2\text{O}_5(g)$ in the container is measured over a period of time, and the measurements are recorded in the following table.

Time (hr)	$[\text{N}_2\text{O}_5](M)$
0	0.160
1.67	0.0800
3.33	0.0400
5.00	0.0200

- (a) Determine the value of the rate constant, k , for the reaction. Include units in your answer.

- (b) The following mechanism is proposed for the decomposition of $\text{N}_2\text{O}_5(g)$.



Identify which step of the proposed mechanism (1, 2, or 3) is the rate-determining step. Justify your answer in terms of the rate law given.

- (c) If this experiment was repeated at the same temperature but with twice the initial concentration of N_2O_5 , would the value of k increase, decrease, or remain the same? Explain your reasoning.

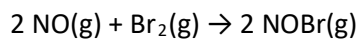
Unit 5: AP Free Response Practice #2 [4 points]

2. The rate of decomposition of azomethane ($\text{C}_2\text{H}_6\text{N}_2$) was studied by monitoring the partial pressure of the reactant as a function of time, and the data below was collected.

Time (s)	P (mmHg)	ln(P)
0	284	5.65
100	220	5.39
150	193	5.26
200	170	5.14
250	150	5.01

- a. Determine if the data above support a first order reaction. Justify your answer. [1 point]
- b. Calculate the rate constant for the reaction, including units. [2 points]
- c. In a second trial, the initial pressure of azomethane was doubled. Does the rate constant, k , for the reaction increase, decrease, or stay the same? Justify your answer. [1 point]

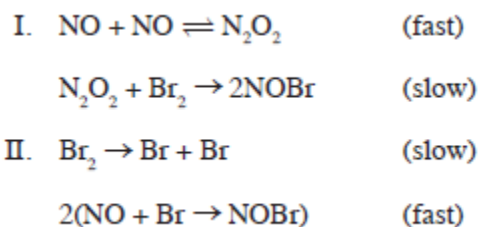
Unit 5: AP Free Response Practice #3 [8 points]



1. The following results were obtained in experiments designed to study the rate of the reaction above.

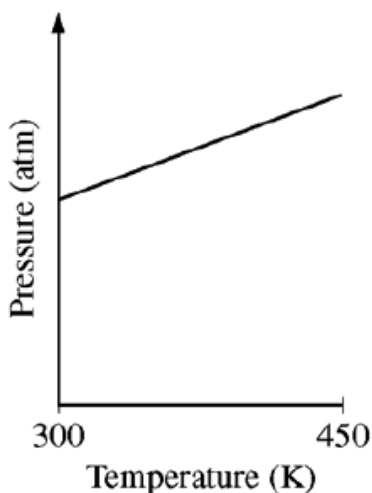
Experiment	Initial Concentration (mol/L)		Initial Rate of Appearance of NOBr (M/sec)
	[NO]	[Br ₂]	
1	0.02	0.02	9.6×10^{-2}
2	0.04	0.02	3.8×10^{-1}
3	0.02	0.04	1.9×10^{-1}

- a. Write the rate law for the reaction. Justify. [3 points]
- b. Calculate the value of the rate constant, k , for the reaction. Include units. [2 points]
- c. Determine the initial rate of the reaction in experiment 1. [1 point]
- d. Which of the following reaction mechanisms is consistent with the rate law established in (a)? Explain your choice. [2 points]



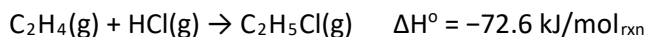
Unit 5: AP Free Response Practice #4 [2013 #5, 8 points]

4. A sample of $\text{C}_2\text{H}_4(g)$ is placed in a previously evacuated, rigid 2.0 L container and heated from 300 K to 450 K. The pressure of the sample is measured and plotted in the graph below.



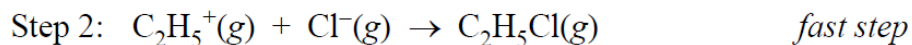
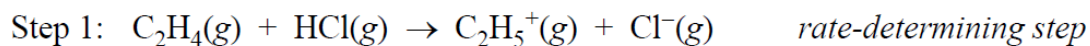
- a. Describe TWO reasons why the pressure changes as the temperature of the $\text{C}_2\text{H}_4(g)$ increases. Your descriptions must be in terms of what occurs at the molecular level. [2 points]

$\text{C}_2\text{H}_4(g)$ reacts readily with $\text{HCl}(g)$ to produce $\text{C}_2\text{H}_5\text{Cl}(g)$, as represented by the following equation.



- b. When $\text{HCl}(g)$ is injected into the container of $\text{C}_2\text{H}_4(g)$ at 450 K, the total pressure increases. Then, as the reaction proceeds at 450 K, the total pressure decreases. Explain this decrease in total pressure in terms of what occurs at the molecular level. [1 point]

It is proposed that the formation of $\text{C}_2\text{H}_5\text{Cl}(\text{g})$ proceeds via the following two-step reaction mechanism.



- c. Write the rate law for the reaction that is consistent with the reaction mechanism above. [1 point]
- d. Identify an intermediate in the reaction mechanism above. [1 point]
- e. Using the axes provided below, draw a curve that shows the energy changes that occur during the progress of the reaction. The curve should illustrate both the proposed two-step mechanism and the enthalpy change of the reaction. [2 points]



- f. On the diagram above, clearly indicate the activation energy, E_a , for the rate-determining step in the reaction. [1 point]